

Kamran Heydarov

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SUMMARY

Chemoinformatics M.Sc. student working at the interface of physical chemistry, molecular modelling and machine learning. Experienced across the full QSPR workflow: curating experimental data from chemical databases, computing and selecting molecular descriptors, training and validating ensemble models, and deploying them as documented tools that non-specialists can use. Author and maintainer of open-source scientific Python libraries for ionic liquids and thermophysical data (pyionics, ILThermoGUI, ILQSPR), with RDKit, Mordred, scikit-learn and XGBoost, alongside quantum chemistry coursework and DFT calculations. Interested in data-driven design of ionic liquids and sustainable solvents.

EDUCATION

University of Strasbourg

M.Sc. in Chemoinformatics

Baku, Azerbaijan

Expected Jun 2027

Azerbaijan State Oil and Industry University

B.Sc. in Chemical Engineering, Chemical Technologies Faculty

Baku, Azerbaijan

Jun 2025

- Graduation thesis, Department of Chemistry and Inorganic Substances Technology ([ResearchGate](#)).

WORK EXPERIENCE

Chemoinformatics and Computational Chemistry Research Group, ASOIU

Baku, Azerbaijan

M1 Chemoinformatics research intern (UFAZ / University of Strasbourg), supervised by Dr. Shamkhal Baybekov

Mar – Jun 2026

- Chemical Space Analysis of Ionic Liquids*: curated the NIST ILThermo database into three clean datasets (5,136 density, 1,524 refractive index, 868 melting point compounds) through deduplication, pure-compound filtering, ambient-condition windows and RDKit standardisation.
- Characterised each compound with 1,613 Mordred 2D descriptors and profiled structural composition using Bemis–Murcko scaffolds and an 88-pattern SMARTS catalogue of ionic-liquid cation and anion classes.
- Projected the filtered descriptor space with UMAP under three hyperparameter configurations and built an interactive Plotly/SMILESDrawer dashboard of the structural and property landscapes.
- Showed the density and refractive-index sets to be over 92% near-duplicates while the melting-point set is structurally distinct, and recommended scaffold- or cluster-based splits over random ones for future QSPR validation.

Azerbaijan State Oil and Industry University

Baku, Azerbaijan

Django Developer & Chief Specialist

Sep 2025 – Jun 2026

- Engineered Django applications and managed database schemas (e.g., MongoDB) to handle critical university data.
- Built and maintain the university's scholarly publishing platforms, front end and back end: press.asoiu.edu.az and the OJS journal system at ojs.asoiu.edu.az; registered DOIs and deposited metadata through Crossref to meet international indexing requirements.
- Coordinated the migration of *Baku Mathematical Journal* to Elsevier Digital Commons across nine international working meetings with the Elsevier team, covering site structure, metadata preparation and article migration; serve as Executive Editor of the journal.
- Designed and programmed the official website of the MacroFrontiers 2026 conference (asoiu.edu.az/mpmc2026), and built the submission, tracking and plagiarism-screening system for the NASCO international conference.
- Administered Elsevier platforms (Scopus, ScienceDirect, SciVal, Reaxys) and Turnitin for ASOIU and UFAZ, and delivered SciVal and Reaxys training seminars.

Azerbaijan State Oil and Industry University

Baku, Azerbaijan

Data-Oriented Lab Assistant, supervised by Dr. Leyla Vazirova

Sep 2023 – Feb 2025

- Assisted professors in undergraduate laboratories at UFAZ and ASOIU.
- Guided 20+ students per lab session, improving their understanding of experimental concepts and helping boost lab grades.
- Taught students statistical methods for chemical analysis, enhancing their data interpretation skills and analytical reasoning.

PROJECT EXPERIENCE

AI-Driven Modeling of Geochemical Element Migration in Oil-Contaminated Soils

UFAZ, Baku

Project member, UFAZ Research Call 2026–2027

2026 – 2027

- Models the migration of geochemical elements, heavy metals and polycyclic aromatic hydrocarbons through soils of Azerbaijan's oil-producing regions, to support remediation planning.
- Lead responsibility for developing and optimising the machine learning models, the laboratory analysis of the organic and inorganic composition of soil samples, and the processing and statistical analysis of the resulting data.
- Combines Random Forest and Gradient Boosting with hybrid neural networks that incorporate geochemical domain knowledge, trained on samples collected across the Absheron Peninsula.
- Planned outputs: validated predictive models, an open geochemical database, and decision-support maps, with partner researchers in France (external member Dr. Hugues Paucot, ICP-MS and elemental analysis).

ILQSPR: QSPR Modelling of IL–Solvent Mixtures (code, app, docs) Baku, Azerbaijan
Data Scientist, joint work with Dr. Shamkhal Baybekov (email) 2025 – Present

- Built QSPR models predicting the density and refractive index of pure ionic liquids and their binary mixtures with water, ethanol, and isopropanol as a function of temperature and IL mole fraction.
- Curated training data from the NIST ILThermo database using pyionics, applying duplicate removal, consistency checks, and structure standardisation.
- Represented ionic liquids with Mordred 2D descriptors computed separately for cation and anion, followed by variance- and correlation-based feature selection.
- Trained consensus ensembles of five XGBoost models using group-based 5-fold cross-validation on IL SMILES, reaching $R^2 = 0.90\text{--}0.96$ across all eight property–solvent systems; manuscript in preparation.
- Deployed the models as a public Streamlit application with online documentation, so that non-specialists can obtain predictions directly from SMILES, temperature and composition.

pyionics Python Library (link) Baku, Azerbaijan
Owner & Developer Mar 2025 – Present

- Developed a Python-based tool for thermodynamic property estimation of ionic liquids and mixtures.
- Integrated models for phase equilibrium and heat capacity prediction to support research and education.
- Optimised the codebase for usability in academic and industrial thermodynamic analysis.

marimo4obs, Obsidian Plugin (code) Baku, Azerbaijan
Owner & Developer 2026

- Built an open-source Obsidian plugin that registers the .py extension and opens marimo reactive Python notebooks inside the vault, each backed by a local marimo server, in both editor and app mode.
- Installs marimo automatically when missing; ships as a community plugin with packaged releases and documented installation.

Chron Shift, Browser Game (play) Baku, Azerbaijan
Creator & Developer 2026

- Developed a single- and multiplayer browser platform game with running and fighting mechanics.

ChemGraph, Process Simulation Software Baku, Azerbaijan
Executive Director Feb 2025 – 2026

- Led the development of ChemGraph, an open-source PFD design simulator supported by Azerbaijan State Oil and Industry University, and extended it for use in education, research and industrial process simulation.

ILThermoGUI Baku, Azerbaijan
Data Scientist, group of Dr. Shamkhal Baybekov (email) Sep 2024 – 2026

- Developed a Python library for installing and accessing NIST ILThermo data, and cleaned and structured the datasets for analysis.
- Designed visualisations and statistical analyses to interpret thermophysical data.

MentorCorp Baku, Azerbaijan
Co-Founder & CTO Jan 2024 – 2026

- Developed a user-friendly interface for the mentoring platform, streamlining resource access and reducing average onboarding time by 30% for new users.

YOUNGH Baku, Azerbaijan
Founder & CEO Jan 2024 – Sep 2024

- Built a Q&A platform with 200+ active users, enabling students to earn up to \$500 collectively in its first six months; sold the platform to a company in our competition.

CONFERENCES & PRESENTATIONS

Poster S. Baybekov, K. Heydarov, O. Abdullayev, Kh. Gaybaliyeva, *QSPR Models for Predicting Density and Refractive Index of Ionic Liquid Binary Mixtures (abstract)*.
 Strasbourg Summer School in Chemoinformatics (CS3-2026), University of Strasbourg, France (35 h) 22–26 Jun 2026

LEADERSHIP & OUTREACH

Science Club Baku, Azerbaijan
Founder 2023 – 2026

- Founded and ran a student science club running academic competitions and outreach events: Integration Bee 2023 Azerbaijan (integral calculus), AnimAI (prompt engineering and generative images), and GenerO with the UFAZ Chemical Engineers Club, on oxygen synthesis in the Martian atmosphere.

SPE Azerbaijan Section Baku, Azerbaijan
Content Writer, SPE Energy School Project (volunteer); letter of appreciation from the Section chair 2024

AWARDS & PROGRAMS

Xsolla–IDDA Baku Game Incubator top 15 studios, with Chron Shift 2026
SABAH.HUB Incubation Project Owner, YOUNGH Jan 2024 – May 2024
UFAZ IGNITE Incubation Program Project Owner, YOUNGH Oct 2023 – Jul 2024
“Ideathon for People” 3rd place, DSC ASOIU / DSC BEU Mar 2021

SKILLS & LANGUAGES

Cheminformatics & ML RDKit, Mordred descriptors, scikit-learn, XGBoost, QSPR modelling, KNIME

Computational chemistry Density Functional Theory for molecular quantum dynamics, thermodynamic property estimation

Programming Python (pandas, NumPy, matplotlib, tkinter), Django, Streamlit, JavaScript, MongoDB, Git, ~~Blender~~ $\LaTeX$

Other tools MATLAB, SolidWorks, Excel

Languages Azerbaijani (native), Turkish (native), English (fluent), Russian (B1), French (A2)

CERTIFICATIONS

Data & programming Data Manipulation with pandas, MongoDB Python Developer Path, Introduction to MongoDB, Intermediate Python, JavaScript (Intermediate and Basic), CSS, GitHub Foundations, KNIME and Excel Fundamentals

AICHE / SACHE[®] Managing Lab Process Safety 1 & 2 and Lab Safety Certificate Program (ELA982, ELA986, ELA954); Fundamentals of Climate Change and Mitigation (ELA137)